

Minimum Wages and Workers by Education Level: Evidence from Peru’s 2022 Reform*

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Abstract

Whether a minimum wage helps the workers it targets depends on how many of them it reaches. I study Peru’s May 2022 increase in the national minimum wage, from 930 to 1,025 soles. Because Peru sets a single national minimum wage, the reform reaches every worker at the same time, and identification comes from differences in exposure by education rather than from staggered timing: low-education workers cluster at the wage floor and are bound by it, while high-education workers earn well above it and provide a lower-exposure comparison group. Using sixteen quarters of the Peruvian national household survey (ENAH, 2021–2024) and a difference-in-differences event study, I find that the reform did not raise the real hourly wages of low-education workers relative to high-education workers. This wage null is precise, holds across the wage distribution, and reappears around the January 2025 increase. Around 2022 I also find a shift from formal employment toward self-employment, concentrated among youth and women, but this margin is less robust and does not replicate in 2025. In a labor market with pervasive informality and weak enforcement, the minimum wage reallocates low-skilled workers across employment arrangements more than it raises their pay, a pattern that qualifies the “lighthouse” view of the Peruvian minimum wage found for earlier reforms.

Keywords: minimum wage; informality; education; difference-in-differences; Peru.

JEL codes: J23, J31, J38, J46, O17.

Highlights

- Effects of Peru’s 2022 minimum wage identified from exposure gaps by education.
- Real wages of low-education workers did not rise relative to high-education workers.

*This paper uses the public-use quarterly employment modules of the Encuesta Nacional de Hogares (ENAH) of the Instituto Nacional de Estadística e Informática (INEI). All estimates use survey weights; the replication package is available in the accompanying repository. All errors are the author’s own.

- The precise wage null reappears around the January 2025 minimum wage increase.
- Formal employment fell and self-employment rose in 2022, but not robustly.
- With high informality, the wage floor reallocates workers more than it lifts pay.

1 Introduction

Whether a minimum wage helps the workers it targets depends on how many of them it actually reaches. In developing countries a large share of employment lies outside the reach of labor regulation, so a statutory wage floor can raise pay for covered workers, displace some of them into the uncovered informal sector, or act only as a reference point for wages that formal firms would have paid in any case. Which of these forces dominates is an empirical question, and its answer turns on how binding the floor is, how well it is enforced, and how freely workers move between the formal and informal margins. I study these questions for Peru, a country that pairs a minimum wage high relative to median pay with one of the highest rates of labor informality in Latin America.

I exploit the increase in Peru’s national minimum wage (the *Remuneración Mínima Vital*, or RMV) from 930 to 1,025 soles, enacted by Supreme Decree 003-2022-TR and effective on 1 May 2022. This 10.2 percent nominal increase was the first change to the RMV in four years, and it occurred as the labor market was emerging from the pandemic. A central institutional fact shapes the analysis: Peru sets a single national minimum wage, with no regional or sectoral variation in the general regime. There is therefore no cross-sectional variation in the level of the policy and no staggered adoption to exploit. As Jiménez (2023) notes for the 2016 reform, credible identification must instead come from differential *exposure* to a policy that changes for everyone at the same time.

My source of differential exposure is education. In the pre-reform quarters the median monthly wage of a dependent employee in Peru was almost exactly 930 soles, the old minimum, so the typical low-education worker sat right at the wage floor, and a ten percent increase in that floor is highly binding. High-education workers, by contrast, have a median wage well above the minimum and are far less likely to be bound by it, although some remain exposed through informal or own-account work. This gap in exposure, rather than a claim of zero exposure for the comparison group, motivates a difference-in-differences design that contrasts low-education workers (those with at most complete secondary schooling, about two thirds of the working-age population) with high-education workers (those with any tertiary schooling). The design applies a long tradition of measuring minimum wage effects from variation in the “bite” of the policy across groups or regions (Card, 1992; Lee, 1999; Machin et al., 2003) to a setting with a single national reform and rich, quarterly household data.

I use the public-use quarterly employment modules of the Encuesta Nacional de Hogares (ENAH) from 2021Q1 through 2024Q4, sixteen quarters that bracket the reform with five pre-reform and eleven post-reform periods. The data are repeated cross-sections, and their

quarterly frequency lets me estimate a dynamic event study, test for differential pre-trends by education, and trace the effect for more than two years after the reform. I examine a broad set of outcomes: real hourly wages and monthly earnings, the probability of employment and of labor force participation, the incidence of formal versus informal employment, hours of work, and the split of employment between wage work and self-employment. Throughout I weight by the survey weights and cluster inference by department, and because treatment is assigned at a coarse level I complement conventional cluster-robust standard errors with the small-sample correction of Pustejovsky and Tipton (2018) and with randomization inference.

Three findings emerge. First, and most robustly, the reform did not raise the real wages of low-education workers relative to high-education workers. The effect on the log real hourly wage is a precisely estimated null, the event study shows flat pre-trends and no break at the reform, and the null holds across the wage distribution. Second, around 2022 the reform is associated with a decline in the probability that a low-education worker holds formal salaried employment and with an offsetting rise in self-employment, a pattern that points to reallocation from the formal to the informal margin rather than to compression of the formal wage distribution. Third, the employment rate of low-education workers falls modestly, but this margin is the least robust and is sensitive to the differential pandemic recovery of early 2021. For the wage and formality results pre-trends are statistically flat; for employment and self-employment they are not, so I bound their conclusions with the sensitivity analysis of Rambachan and Roth (2023) and a battery of specification and placebo checks. I then replicate the whole design on the January 2025 reform, which raised the RMV to 1,130 soles in a cleaner, post-pandemic window. The wage null reappears; the compositional effects do not, which leads me to read them as specific to the 2022 episode.

The contribution is a combination rather than a new tool, and I state it without over-claiming. This is the first evaluation of the 2022 Peruvian minimum wage reform and, to my knowledge, the first study of any Peruvian reform using the post-pandemic quarterly ENAHO; the existing micro evidence stops at the 2016 reform (Jiménez, 2023) or at the reforms of the early 2000s (Jaramillo and López, 2006; Céspedes, 2005). It is also the first to make heterogeneity by education the central object of a worker-level event study of the Peruvian minimum wage, where prior work cuts heterogeneity by income range or age or constructs a department-level dose of exposure. And it places the informal margin at the center of the analysis, showing that adjustment runs through the composition of employment rather than through pay. I claim no novelty for the identification idea, which is the standard exposure or bite design, nor for the econometric toolkit; the value added is the pairing of a new reform and a policy-relevant post-pandemic period with an education-based worker-level design and a careful treatment of informality and of the threats to identification that such a

design entails.

The rest of the paper proceeds as follows. Section 2 describes the Peruvian minimum wage and the reform, and Section 3 places the paper in the literature. Section 4 presents the data and the construction of the key variables, and Section 5 sets out the empirical strategy and the identifying assumptions. Section 6 reports the main results, Section 7 the robustness and inference checks, and Section 8 the heterogeneity and distributional analysis. Section 9 discusses mechanisms, external validity, and limitations, and Section 10 concludes.

2 Institutional background

2.1 The Peruvian minimum wage

Peru sets a single national minimum wage, the *Remuneración Mínima Vital* (RMV), which applies to workers in the private-sector labor regime. Unlike federal systems such as the United States or Brazil, and unlike countries with occupational or regional wage floors, Peru has no sub-national or industry-specific statutory minimum in the general regime. A small number of activities carry fixed multiples of the RMV (for example the mining premium and the agricultural daily wage), but these are pegged mechanically to the national value and move with it. The consequence for research design is decisive: because the level of the RMV is common to the whole country, there is no cross-sectional variation in the policy and no staggered timing of adoption. Any credible research design must therefore exploit differences in how exposed workers are to a change that reaches everyone simultaneously.

In principle the RMV is to be adjusted periodically by the tripartite National Labor Council in line with inflation and productivity. In practice the Council has never set the RMV, and the level is fixed at the discretion of the executive through a Supreme Decree. Adjustments are consequently infrequent and irregular in both timing and magnitude, and Peru is among the most volatile minimum wage setters in the region (Jiménez, 2023; Céspedes, 2005). The RMV had stood at 930 soles since April 2018.

Two features make the RMV unusually binding by international standards. First, its ratio to the median wage is high. In my pre-reform sample the national median monthly wage of a dependent employee is almost exactly 930 soles, so the old minimum already sat at the median of the covered wage distribution and the Kaitz index, the ratio of the minimum to the median wage, is close to one nationally and well above one in the poorer departments. Second, compliance is far from universal. Consistent with earlier work (Jaramillo and López, 2006; Céspedes, 2005), a large share of workers earn below the statutory floor, especially in rural areas, in small firms, and in the informal sector, while compliance inside registered

formal firms is high. Enforcement capacity is limited. These features imply that a nominal increase in the RMV can bind mechanically for formal workers near the floor while leaving much of the informal sector formally uncovered, so that the policy’s real reach depends on the degree of informality and on the ease of movement across the formal and informal margins.

2.2 The 2022 reform

The reform I study was enacted by Supreme Decree 003-2022-TR, signed on 2 April 2022 and published on 3 April, and took effect on 1 May 2022. It raised the RMV from 930 to 1,025 soles, an increase of 95 soles or 10.2 percent, the first change in four years. Figure 1 plots the nominal and real value of the RMV over 2021–2025. Two points are worth noting. The nominal increase of 10.2 percent was partly offset by inflation, which ran at roughly 8 percent over 2022, so the increase in the real value of the floor was more modest and eroded over the following two years. This tempers the expected size of any wage response and is relevant when comparing my estimates with those from lower-inflation episodes. The reform also occurred during the post-pandemic recovery, a period of unusual labor market turbulence that I address explicitly in the empirical analysis.

A second reform, enacted by Supreme Decree 006-2024-TR and effective on 1 January 2025, raised the RMV further to 1,130 soles. This change falls just outside my main study window, but the ENAHO quarters for 2025 are available and I use this second reform as an out-of-sample replication of the design in Section 7. The two reforms are of very similar magnitude, 10.2 percent each, which makes the 2025 episode a natural validation exercise in a calmer, post-pandemic environment.

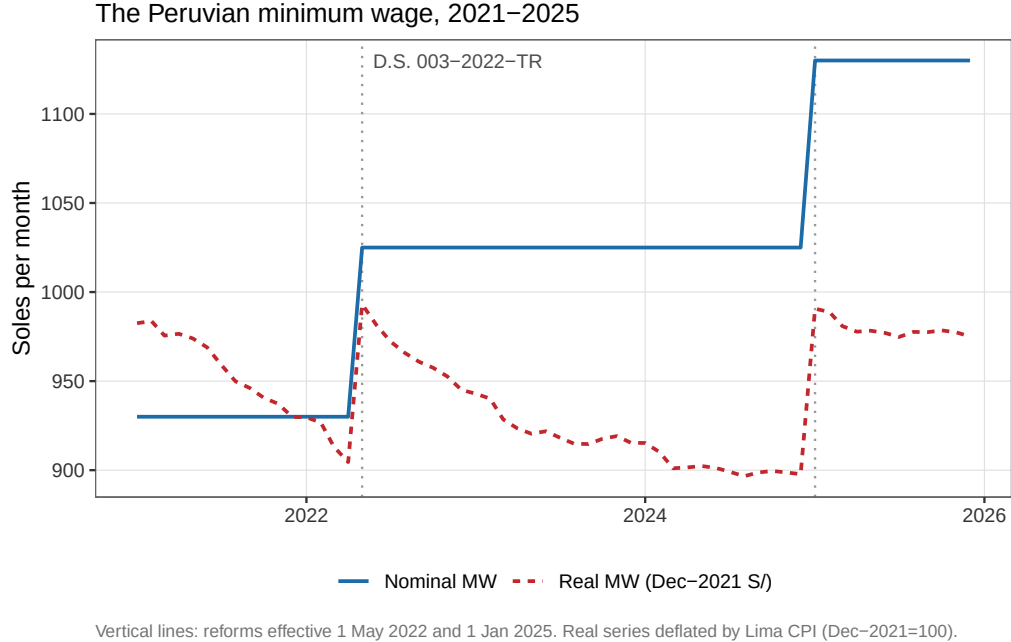


Figure 1: The nominal and real Peruvian minimum wage, 2021–2025. The real series is deflated by the Lima consumer price index (December 2021 = 100). Vertical dotted lines mark the reforms effective 1 May 2022 (Supreme Decree 003-2022-TR) and 1 January 2025 (Supreme Decree 006-2024-TR).

3 Related literature

My paper connects three strands of research: the theory and evidence on minimum wage employment effects, the evidence for developing countries with large informal sectors, and the specific literature on Peru.

Theory and the canonical evidence. The competitive model predicts that a binding wage floor reduces employment among low-productivity workers, with an effect whose size depends on labor demand elasticities and on the fraction of workers bound by the floor (Brown, 1999). Models with employer wage-setting power deliver different predictions: under monopsony a moderate minimum wage can raise employment, and it typically compresses the lower tail of the wage distribution (Manning, 2003). The empirical literature has moved from a presumption of sizable disemployment toward a consensus of small effects on the margin studied. The case study of Card and Krueger (1994) and the broader synthesis in Card and Krueger (1995) found little evidence of job loss from the New Jersey reform, and the border-discontinuity design of Dube et al. (2010) and the bunching approach of Cengiz et al. (2019) likewise find that recent United States increases raised wages at the bottom

with limited employment effects. Harasztosi and Lindner (2019) show that a large Hungarian increase was absorbed mostly through prices rather than employment. A recurring theme, which I build on directly, is that effects should be measured where the minimum “bites.” Card (1992) uses regional variation in the fraction of workers affected, Lee (1999) uses the gap between the minimum and the median wage, and Machin et al. (2003) exploit a low-wage sector in which the introduction of the United Kingdom minimum was especially binding. My education-based design is an application of the same logic to a single national reform.

Developing countries and informality. In economies with large informal sectors the minimum wage interacts with the boundary between covered and uncovered work, and two opposing forces appear repeatedly. A “lighthouse” effect can raise wages even in the informal sector, where the minimum serves as a norm or reference (Lemos, 2009; Bosch and Manacorda, 2010), while a reallocation effect can push workers out of formal employment and into informal or self-employed arrangements (Ham, 2018). The Latin American evidence spans this range. Studies of Brazil find wage compression with small employment effects and evidence of spillovers into the informal sector (Lemos, 2009, 2004). Work on Ecuador finds positive wage effects concentrated among low-income workers with heterogeneous hours responses (Wong, 2019). The Honduran case is the clearest counterexample to the lighthouse view: Ham (2018) documents that a high minimum wage reduced covered employment and shifted workers toward informality, with no reduction in poverty. For Uruguay, Katzkowicz et al. (2021) use a density-discontinuity approach in the domestic-work sector to show that effects concentrate among the least skilled. The common methodological thread is the use of exposure or bite measures, either geographic Kaitz indices or fraction-affected shares, and a careful accounting of the informal margin, both of which I adopt.

Peru. The domestic literature is thin and, until recently, dated. Early studies of the reforms of the early 2000s find small and often statistically weak effects. Jaramillo and López (2006) analyze the 2003 increase using short quarterly panels for metropolitan Lima and conclude that the reform was largely innocuous, with negative employment-retention effects concentrated near the floor and evidence of displacement from the formal to the informal sector. Céspedes (2005) estimates a weakly significant employment elasticity of about -0.13 and shows that the RMV Granger-causes formal wages, an early statement of the lighthouse view for Peru. The most credible modern study is Jiménez (2023), who evaluates the 2016 reform using a difference-in-differences design with a department-level treatment dose, defined as the pre-reform share of formal workers earning between the old and new minimum. That study finds no disemployment, a rise in formality of about 2.5 to

2.8 percentage points, and wage gains of roughly 4 to 5 percent, and it interprets the pattern through employer market power. My paper differs from this precedent in three respects that I make explicit. I study a later reform and the post-pandemic period; I define exposure at the worker level through education rather than at the department level; and I find a different pattern of adjustment, with no wage gain and a decline rather than an increase in formality. I return to the reconciliation of these findings in Section 9. As Jiménez (2023) emphasizes, the binding constraint for any Peruvian design is the absence of sub-national variation in the nominal minimum, which forces the researcher to manufacture differential exposure; my contribution is to do so through the education dimension and to trace the consequences for the formal and informal margins.

Econometrics of difference-in-differences. Methodologically I rely on the modern difference-in-differences literature. Because my reform occurs at a single date, the recent concerns about negative weighting and forbidden comparisons under staggered adoption (Goodman-Bacon, 2021; de Chaisemartin and D’Haultfoeuille, 2020; Sun and Abraham, 2021; Callaway and Sant’Anna, 2021) do not bind: with a common treatment date the two-way fixed-effects estimator and its event-study analog are unbiased for a convex average of group-time effects (Roth et al., 2023; Baker et al., 2022). I nonetheless use the doubly robust estimator of Sant’Anna and Zhao (2020) as a building block, and I take the threat to identification, namely differential trends by skill, seriously through the sensitivity analysis of Rambachan and Roth (2023). For inference with few clusters I follow Bertrand et al. (2004), Cameron et al. (2008), and Pustejovsky and Tipton (2018). Distributional effects are estimated with the unconditional quantile method of Firpo et al. (2009).

4 Data and measurement

4.1 The ENAHO quarterly employment modules

I use the public-use quarterly files of the Encuesta Nacional de Hogares (ENAHO), the national household survey conducted by Peru’s Instituto Nacional de Estadística e Informática (INEI). ENAHO is a nationally representative, stratified, multi-stage survey with continuous fieldwork; the quarterly employment and income module (module 500) records detailed labor market information for each household member of working age. I pool the sixteen quarters from the first quarter of 2021 through the fourth quarter of 2024, which yields about 347,000 person-quarter records of working-age individuals. I add the four quarters of 2025 for the out-of-sample validation exercise. All statistics use the module’s sampling weights, and I

cluster inference by department.

I restrict the sample to individuals of working age, defined as ages 14 to 65, and I drop observations with missing education. My main wage analysis is further restricted to dependent wage employees, for whom the minimum wage is the relevant statutory floor, while the employment, participation, formality, and self-employment outcomes are defined over the working-age population or over the employed, as appropriate.

4.2 Education and the definition of skill groups

The treatment and comparison groups are defined by educational attainment, recorded in ENAHO variable `p301a`, the highest level of schooling completed. I validated the coding of this variable against the weighted distribution of attainment in the data; it corresponds to the standard INEI classification, ranging from no schooling through postgraduate study. I define *low-skilled* workers as those with at most complete secondary schooling (no education, initial, primary, or secondary), and *high-skilled* workers as those with any tertiary schooling (non-university higher, university, or postgraduate). I exclude the small residual category of special education, which is not ordered on the skill dimension, and observations with missing education.

This partition has three attractive features. First, it is a bright line in the Peruvian labor market: completion of tertiary schooling is the main credential that separates workers who compete for jobs paying near the minimum wage from those who do not. Second, it splits the working-age population into groups of comparable size, roughly two thirds low-skilled and one third high-skilled, which gives adequate statistical power on both sides. Third, and most important for identification, the two groups differ sharply in their exposure to the minimum wage. As Table 1 shows, in the pre-reform period the median monthly wage of a low-skilled dependent employee is close to the statutory floor, whereas the median high-skilled employee earns well above it. The classification follows a long line of work that treats education as a primary determinant of where the minimum wage bites (Lemos, 2009; Katzkowicz et al., 2021).

4.3 Outcome variables

I construct the following outcomes. *Real hourly wage* is monthly labor income in the main occupation divided by monthly hours, deflated by the Lima consumer price index to December 2021 soles; I build monthly labor income following the ENAHO convention used by Jiménez (2023), and I winsorize the wage at the first and last half-percentiles to limit

the influence of outliers. *Real monthly earnings* is the same numerator without the hours adjustment. *Employment* is an indicator for being employed in the reference week, defined over the working-age population, and *labor force participation* is an indicator for being employed or unemployed. *Formal employment* is an indicator for holding a formal job among the employed. *Self-employment* is an indicator for own-account work among the employed, *weekly hours* is hours worked in the main occupation, and *paid below the minimum* is an indicator that monthly earnings fall below the statutory floor, a direct measure of the bite of and compliance with the policy.

A measurement issue arises for informality. INEI’s official informal-employment indicator, *ocupinf*, is present in the 2021 through 2023 files but is not released in the 2024 and 2025 files. Because my study window extends through 2024, I reconstruct a consistent informality indicator from underlying primitives that are available in every year: pension affiliation, the possession of a formal labor contract, the registration status of the productive unit, and the category of employment. I validate the reconstruction against the official *ocupinf* in 2021 through 2023, where both are available, and obtain agreement of about 88 percent at the worker level, with an implied informality rate within one percentage point of the official figure. I use the official indicator wherever it exists and the reconstruction only for 2024 and 2025, and I report a robustness check that restricts the formality analysis to the years covered by the official measure. Full details of the construction are in Appendix A.

4.4 Descriptive statistics

Table 1 reports weighted means of the main variables by skill group in the pre-reform period. The two groups differ in the expected ways. Low-skilled workers are older on average, more likely to be male and rural, and have far lower wages. The employment rate is similar across groups, but the composition of employment differs starkly: low-skilled workers are much more likely to be informal and self-employed, and far more likely to earn below the statutory minimum. Roughly seven in ten low-skilled dependent employees earn below the new minimum in the pre-reform period, against fewer than four in ten high-skilled employees, which is the exposure gap on which my design rests. The high informality of the low-skilled group, around 85 percent against about 49 percent for the high-skilled, is central to interpreting the results: for most low-skilled workers the statutory floor is not directly enforced, so the margin along which the labor market can adjust is the allocation of workers across formal and informal arrangements rather than the level of pay within formal jobs.

Table 1: Descriptive statistics by skill group, pre-reform period

	Low-skilled ($\leq$ sec. complete)	High-skilled ($\geq$ higher)	Difference
Age	35.923	34.649	1.275
Female	0.496	0.501	-0.005
Urban	0.762	0.931	-0.169
Married	0.530	0.413	0.117
Years of education	8.573	14.468	-5.895
Labour force participation	0.728	0.799	-0.071
Employed	0.695	0.735	-0.040
Wage employee	0.272	0.475	-0.203
Self-employed	0.392	0.244	0.148
Formal (if employed)	0.129	0.469	-0.340
Informal (if employed)	0.871	0.531	0.340
Weekly hours (if employed)	39.737	41.438	-1.701
Real hourly wage (S/)	3.556	10.569	-7.013
Real monthly earnings (S/)	703.888	1,667.342	-963.454
Below MW (if employed)	0.718	0.360	0.358
Observations	65,954	27,278	

Notes: Weighted means using ENAHO survey weights, working-age individuals (14–65) in the pre-reform quarters (2021Q1–2022Q1). Low-skilled: education at most complete secondary ($\text{p301a} \leq 6$). High-skilled: any tertiary education ($\text{p301a} \geq 7$). Wage and hours variables are conditional on employment; the hourly wage is further conditional on being a dependent employee.

Figure 2 plots the weighted quarterly means of the main outcomes by skill group over the study window, with the reform marked. The raw series already hint at the results that follow: the wage series of the two groups move in parallel throughout, while the formality and self-employment series begin to diverge around the reform. These are unconditional means; the regression analysis that follows adjusts for composition and seasonality.

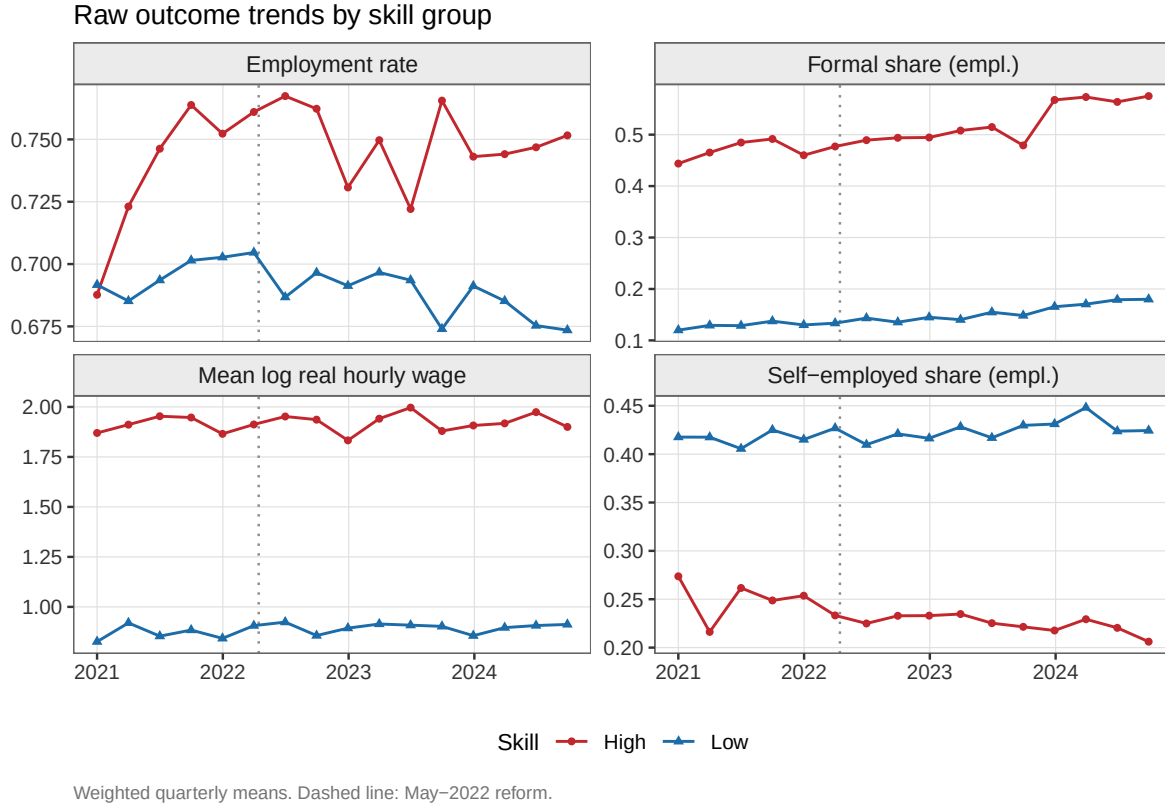


Figure 2: Raw outcome trends by skill group, 2021–2024. Weighted quarterly means for low- and high-skilled workers. The dashed vertical line marks the May 2022 reform.

Figure 3 displays the regional variation in the bite of the minimum wage, measured by the Kaitz index, the ratio of the new minimum to the pre-reform median dependent wage in each department. The index ranges from about 0.79 in the richest department to about 2.79 in the poorest, so that in much of the country the new minimum exceeds the median wage. This variation is predetermined with respect to the reform and provides the basis for the triple-difference and continuous-intensity checks in Section 7.

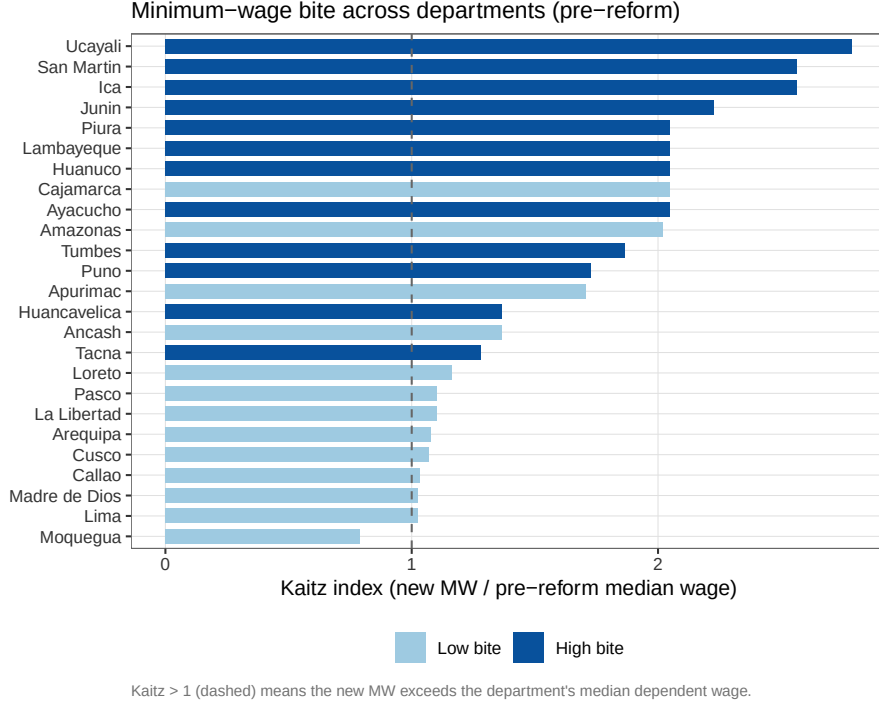


Figure 3: Regional variation in the bite of the minimum wage. The Kaitz index is the ratio of the new minimum wage (1,025 soles) to the department’s median monthly wage of dependent employees in the pre-reform quarters. Departments above the dashed line have a median wage below the new minimum.

5 Empirical strategy

5.1 Potential outcomes and the estimand

Let i index individuals, $g \in \{\text{Low}, \text{High}\}$ their skill group, and t the quarter. The reform becomes effective in the second quarter of 2022, which I denote t^* ; define the post-reform indicator $\text{Post}_t = \mathbf{1}\{t \geq t^*\}$ and the treatment indicator $\text{Low}_i = \mathbf{1}\{g_i = \text{Low}\}$. Following the potential-outcomes framework, let $Y_{it}(1)$ and $Y_{it}(0)$ denote the outcome of individual i in quarter t with and without the reform being in force. The parameter of interest is the average effect of the reform on the low-skilled, relative to the counterfactual in which they had experienced the same evolution as the high-skilled,

$$\text{ATT} = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid \text{Low}_i = 1, t \geq t^*]. \quad (1)$$

The data are repeated cross-sections rather than a panel, so the object is identified from comparisons of group means over time rather than from within-person changes.

Because Peru’s minimum wage is national and changes on a single date, there is no variation in treatment *timing*; every worker is exposed to the same policy at the same moment. Identification instead rests on variation in the *intensity* of exposure across skill groups. The design is therefore a canonical two-group difference-in-differences with many periods, not a staggered-adoption design. This distinction matters for the choice of estimator. The recent econometric literature has shown that the two-way fixed-effects estimator can be badly biased when treatment timing is staggered and effects are heterogeneous (Goodman-Bacon, 2021; de Chaisemartin and D’Haultfoeuille, 2020; Sun and Abraham, 2021; Callaway and Sant’Anna, 2021). With a common treatment date, however, these problems do not arise: the two-way fixed-effects estimator and its event-study analog recover a convex, properly weighted average of treatment effects (Roth et al., 2023; Baker et al., 2022). I nonetheless report the doubly robust estimator of Sant’Anna and Zhao (2020) as a complementary building block.

5.2 Estimating equations

My baseline specification is the two-way fixed-effects difference-in-differences model

$$Y_{idt} = \beta (\text{Low}_i \times \text{Post}_t) + \gamma \text{Low}_i + \lambda_t + \mu_d + X'_{it}\theta + \varepsilon_{idt}, \quad (2)$$

where d denotes the department, λ_t are quarter fixed effects, μ_d are department fixed effects, and X_{it} is a vector of controls comprising age, its square, sex, an urban indicator, marital status, and years of education (which captures variation in attainment within skill groups). The coefficient β is the difference-in-differences estimate. The quarter fixed effects absorb any shock common to both skill groups in a given quarter, including aggregate inflation, so that estimates in logs are invariant to whether wages are expressed in nominal or in nationally deflated real terms. Observations are weighted by the survey weights, and standard errors are clustered by department.

To trace the dynamics of the effect and to test the identifying assumption, I estimate the dynamic event-study specification

$$Y_{idt} = \sum_{k \neq -1} \beta_k (\text{Low}_i \times \mathbf{1}\{t - t^* = k\}) + \gamma \text{Low}_i + \lambda_t + \mu_d + X'_{it}\theta + \varepsilon_{idt}, \quad (3)$$

where k indexes quarters relative to the reform and the quarter immediately before the reform ($k = -1$, the first quarter of 2022) is the omitted reference. The coefficients β_k for $k < 0$ measure differential pre-trends between the skill groups and provide a direct test of

parallel trends; the coefficients for $k \geq 0$ trace the path of the effect.

5.3 Identifying assumptions and threats

Identification of the ATT from equation (2) requires that, absent the reform, the outcomes of low- and high-skilled workers would have followed parallel paths, conditional on the controls and fixed effects. Three threats deserve attention.

First, *differential trends*. Education is correlated with sector, region, and formality, and the two skill groups could have been on diverging paths for reasons unrelated to the minimum wage. I assess this directly through the pre-reform event-study coefficients and a joint test of their equality to zero, and I probe robustness to department-specific linear trends. Because my window opens in the depth of the pandemic recovery, when the employment of low- and high-skilled workers rebounded at different rates, I pay particular attention to the first quarter of 2021 and report specifications that drop it.

Second, *sensitivity to residual violations*. Even when pre-trends are statistically flat, they need not be exactly zero. I therefore report the sensitivity analysis of Rambachan and Roth (2023), which asks how large a post-reform violation of parallel trends, relative to the largest violation observed before the reform, would be needed to overturn each conclusion.

Third, *spillovers and general equilibrium*. If the reform displaces low-skilled workers into the informal sector, it may indirectly depress wages there, and if high-skilled workers are imperfect substitutes their outcomes could also move, contaminating the comparison group. I cannot rule this out, and I interpret the estimates as relative effects. The comparison group is not entirely unexposed, since a minority of high-skilled workers also earn near the floor through informal or own-account work; this imperfect exposure attenuates my estimates toward zero, so the design understates rather than overstates the true effect on the fully exposed. I assess the plausibility of the comparison by showing that the median high-skilled wage sits well above the floor and that the high-skilled event-study path shows no break at the reform.

5.4 Inference

The treatment is assigned at a coarse level, the skill group, which raises the standard concern that conventional standard errors overstate precision (Bertrand et al., 2004). I address this in three ways. I cluster by department, which allows arbitrary within-department correlation across individuals and quarters; I report an alternative that clusters by department and skill group; and I apply the bias-reduced CR2 small-sample correction of Pustejovsky and Tipton

(2018) to department-level collapsed cells, where it is computationally feasible. Finally, because these corrections may still be optimistic with a two-group design, I complement them with randomization inference at the department level, permuting the regional bite of the minimum wage across departments to construct an exact null distribution for the exposure-intensity design.

6 Main results

6.1 The wage response

The central result of the paper is that the 2022 reform did not raise the real wages of low-skilled workers relative to high-skilled workers. Column (1) of Table 2 reports the two-way fixed-effects difference-in-differences estimate from equation (2). The coefficient on the interaction of the low-skill indicator with the post-reform indicator is -0.009 for the log real hourly wage, with a standard error of 0.022, an estimate that is economically small and statistically indistinguishable from zero. The implied point estimate is a relative change of less than one percent, and the ninety-five percent confidence interval rules out relative wage gains larger than about three percent. The result for log monthly earnings is likewise a tight null. The doubly robust estimator of Sant’Anna and Zhao (2020) in column (2) is negative and imprecise but tells the same qualitative story of no positive wage effect.

The dynamic event study confirms that this null is not an artifact of averaging over offsetting periods. The top-left panel of Figure 4 plots the coefficients β_k from equation (3) for the log real hourly wage. The pre-reform coefficients are small and statistically insignificant, and a joint test of their equality to zero does not reject, with a p -value of 0.60 reported in column (3) of Table 2. After the reform the coefficients hover around zero with no discernible break. The wage distribution tells a consistent story: Figure 5 shows the earnings distribution of low-skilled wage employees before and after the reform, with the old and new minima marked. There is a visible mass of workers near the old minimum, but the distribution does not shift appreciably toward the new floor, and a large share of workers remain below it, reflecting the pervasive non-compliance documented in Section 4.

Table 2: Difference-in-differences estimates of the 2022 minimum-wage reform

Outcome	(1) TWFE DiD	(2) Doubly robust	(3) Pre-trend p
Log real hourly wage	-0.009 (0.022)	-0.058 (0.054)	0.597
	$N = 89,419, \bar{y} = 1.361$		
Log real monthly earnings	0.005 (0.019)	-0.034 (0.054)	0.038
	$N = 91,937, \bar{y} = 6.532$		
Employment	-0.025*** (0.005)	0.020 (0.041)	0.000
	$N = 296,954, \bar{y} = 0.708$		
Labour force part.	-0.007 (0.005)	-0.015 (0.036)	—
	$N = 296,954, \bar{y} = 0.745$		
Formal employment	-0.036*** (0.007)	0.009 (0.027)	0.443
	$N = 214,209, \bar{y} = 0.270$		
Self-employment	0.029*** (0.006)	0.053** (0.024)	0.000
	$N = 214,209, \bar{y} = 0.357$		
Weekly hours	0.810*** (0.182)	-0.453 (1.041)	0.109
	$N = 210,534, \bar{y} = 40.709$		
Paid below MW	-0.002 (0.009)	0.003 (0.024)	0.550
	$N = 91,937, \bar{y} = 0.496$		

Notes: Each row is a separate regression. Column (1) reports the two-way fixed-effects DiD coefficient on $\text{Low} \times \text{Post}$ from equation (2), with department and quarter fixed effects and controls (age, age squared, sex, urban, marital status, years of education). Column (2) reports the doubly robust DiD estimator of Sant’Anna and Zhao (2020) collapsing to pre/post and dropping the transition quarter. Column (3) is the p -value of a joint test that the pre-reform event-study interactions are zero. Standard errors clustered by department in parentheses. Survey weights used throughout. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

That the reform failed to lift wages at the bottom is not, in this setting, a surprising result. The nominal increase of ten percent was substantially eroded by inflation of about eight percent over 2022, so the real value of the floor rose only modestly. More fundamentally, most low-skilled workers are informal or self-employed, for whom the statutory minimum is not enforced. For them the minimum wage is at best a norm, and the data suggest it did not operate as a binding one around this reform. The precise null stands in contrast to the “lighthouse” pattern found for the 2016 reform by Jiménez (2023), and I return to this contrast in Section 9.

6.2 Employment and its composition

If the reform did not raise pay, did it affect employment or its composition? Here the evidence is more suggestive than conclusive, and it points to adjustment along the margin between formal and informal work rather than to large changes in the level of employment.

The estimated effect on the probability of formal employment among the employed is -0.036 (column 1 of Table 2), a decline of about three and a half percentage points for low-skilled relative to high-skilled workers, and it is precisely estimated under department clustering. The pre-trend test does not reject parallel trends ($p = 0.44$), and the event study in Figure 4 shows relative formality of the low-skilled drifting down after the reform. Mirroring this, the probability of self-employment rises by about three percentage points, an estimate that is statistically significant and, as shown below, robust to randomization inference. Taken together these two results describe a reallocation of low-skilled workers away from formal salaried jobs and toward own-account work, the classic informalization margin emphasized by Jaramillo and López (2006) for Peru and by Ham (2018) for Honduras. Weekly hours in the main occupation rise modestly, consistent with a shift toward self-employment, where hours are less constrained.

The effect on the overall employment rate is negative, at about -2.5 percentage points, but it is the least robust of my estimates. As the event study makes clear, the employment series carries a differential pre-trend concentrated in the first quarter of 2021, when the labor market was still recovering unevenly from the pandemic across skill groups. Once that quarter is excluded the pre-trend test no longer rejects, but the point estimate also attenuates, and the effect does not survive my more conservative inference procedures. I therefore treat the employment-level result as inconclusive and place little weight on it, reserving the interpretive emphasis for the compositional shift, which is more stable. Finally, the indicator for being paid below the minimum shows no significant change, meaning that the reform did not measurably improve compliance at the bottom of the distribution.

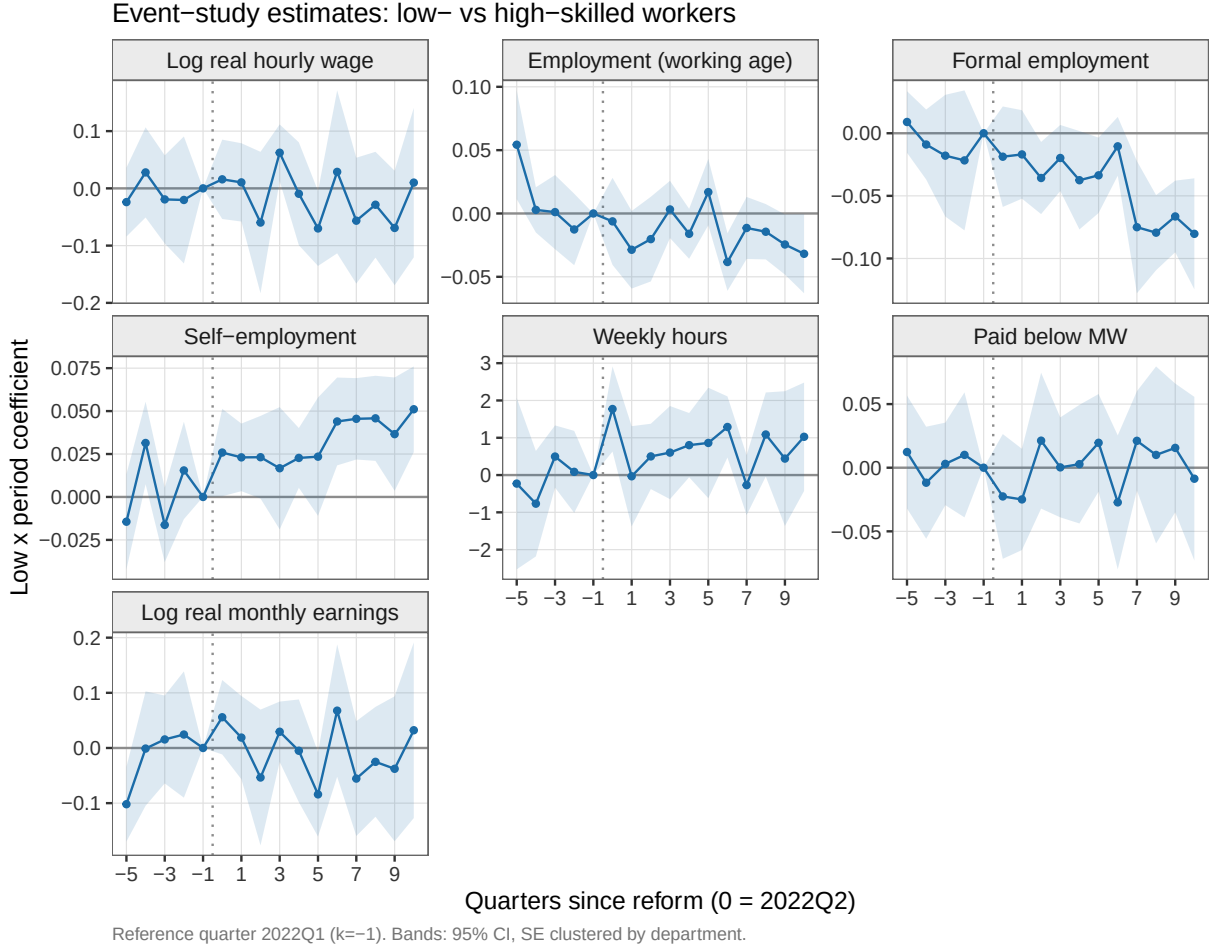


Figure 4: Event-study estimates of the 2022 reform. Each panel plots the coefficients β_k on the interaction of the low-skill indicator with quarter indicators from equation (3), relative to the omitted quarter 2022Q1 ($k = -1$). The reform is effective at $k = 0$ (2022Q2). Shaded bands are ninety-five percent confidence intervals with standard errors clustered by department.

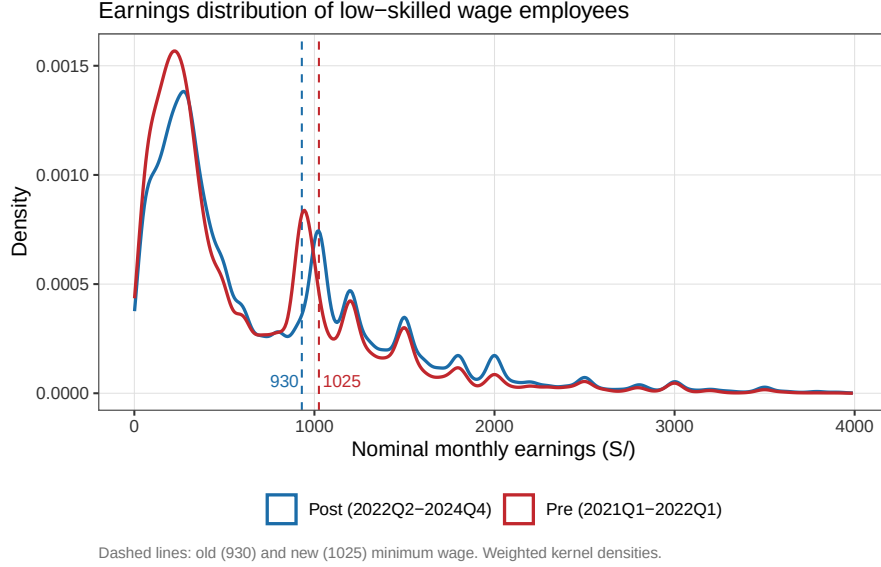


Figure 5: Earnings distribution of low-skilled wage employees, before and after the reform. Weighted kernel densities of nominal monthly earnings, with the old (930) and new (1,025) minimum wage marked by dashed lines.

7 Robustness and inference

This section subjects the main results to a battery of specification, inference, and design checks, and then to an out-of-sample replication. The overall message is that the wage null is very robust, the compositional results are moderately robust, and the employment-level result is fragile.

7.1 Specification robustness

Table 3 reports the $\text{Low} \times \text{Post}$ coefficient for the four headline outcomes under eight alternative specifications: dropping the controls, dropping the pandemic quarter of early 2021, adding department-specific linear trends, restricting to prime-age workers, using an alternative education cutoff that removes the categories at the boundary between the two groups, adding one-digit occupation and industry fixed effects, and clustering by department and skill. The wage estimate remains a tight null across every specification. The formality and self-employment estimates are stable in sign and magnitude throughout, moving within a narrow band. The employment estimate, by contrast, attenuates noticeably when the pandemic quarter is dropped or when department-specific trends are included, which reinforces my reluctance to interpret it causally.

Table 3: Specification robustness of the DiD estimates

Specification	Log hourly wage	Employment	Formal empl.	Self-employment
Baseline	-0.009 (0.022)	-0.025*** (0.005)	-0.036*** (0.007)	0.029*** (0.006)
No controls	0.016 (0.023)	-0.021*** (0.005)	-0.029*** (0.007)	0.034*** (0.006)
Drop 2021Q1 (pandemic)	-0.012 (0.018)	-0.013*** (0.004)	-0.032*** (0.008)	0.025*** (0.005)
Department linear trends	-0.003 (0.022)	-0.017*** (0.005)	-0.034*** (0.007)	0.028*** (0.006)
Prime age (25–55)	-0.014 (0.031)	-0.022*** (0.004)	-0.030*** (0.007)	0.021*** (0.006)
Alternative education cutoff	-0.008 (0.018)	-0.038*** (0.006)	-0.048*** (0.006)	0.041*** (0.007)
Occupation & industry FE	0.000 (0.022)	-0.014*** (0.004)	-0.027*** (0.006)	0.017*** (0.004)
Cluster dept. $\times$ skill	-0.009 (0.026)	-0.025 (0.019)	-0.036*** (0.006)	0.029*** (0.006)

Notes: Each cell is the Low $\times$ Post DiD coefficient (standard error clustered by department below) for the outcome in the column heading, under the specification in the row. The baseline is equation (2). The alternative education cutoff defines low-skilled as at most incomplete secondary and high-skilled as at least complete non-university tertiary, dropping the boundary categories. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

7.2 Inference

Because the treatment is assigned at the level of the skill group, I assess whether the significance of the compositional results survives more demanding inference. Table 4 collects the evidence. Columns (1) through (4) report p -values for the baseline effect under department clustering, department-by-skill clustering, the CR2 small-sample correction of Pustejovsky and Tipton (2018) applied to department-level collapsed cells, and randomization inference that permutes the regional bite of the minimum wage across departments. The self-employment effect is the most resilient: it remains significant under department-by-skill clustering and under randomization inference. The formality effect is significant under both clustering schemes but only marginal under the most conservative CR2 and randomization procedures, which are low-powered with twenty-five departments. The employment effect loses significance under every procedure beyond simple department clustering. The wage null is, of course, unaffected.

Table 4: Inference robustness, design robustness, and out-of-sample validation

Outcome	(1) Dept.	(2) Dept. $\times$ skill	(3) CR2	(4) RI	(5) Continuous	(6) 2025 reform
Log hourly wage	0.704	0.740	0.728	0.310	-0.013	0.027
Employment	0.000	0.213	0.505	0.213	-0.020**	-0.002
Formal empl.	0.000	0.000	0.132	0.234	-0.008***	0.004
Self-employment	0.000	0.000	0.141	0.021	0.010***	-0.014**

Notes: Columns (1)–(4) report p -values for the baseline Low $\times$ Post effect under, respectively, department clustering, department $\times$ skill clustering, the CR2 small-sample correction (Pustejovsky and Tipton, 2018) applied to department-level collapsed cells, and randomization inference permuting the regional bite across departments (2,000 draws). Column (5) reports the continuous-exposure estimate, the coefficient on Post $\times$ (standardized department Kaitz index). Column (6) reports the DiD estimate for the January-2025 reform (window 2024–2025). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

7.3 Design robustness: bite intensity and triple differences

My identification rests on differential exposure by education, but the minimum wage also bites differentially across regions, and this provides an independent source of variation. Column (5) of Table 4 reports a continuous-exposure specification in the spirit of Card (1992), which replaces the binary treatment with the standardized department Kaitz index interacted with the post indicator. The results corroborate the skill-based design: in departments where the minimum wage bites harder, formal employment falls and self-employment rises by significantly more after the reform, while wages do not respond. A triple-difference specification that interacts the skill contrast with an indicator for high-bite departments is not statistically significant, which I read as a lack of power in the binary regional split rather than as evidence against the mechanism, given that the continuous version is informative.

7.4 Sensitivity to violations of parallel trends

Even flat pre-trends are estimated with error, so I apply the sensitivity analysis of Rambachan and Roth (2023). Figure 6 plots the robust confidence set for the average post-reform effect on employment and formality as I allow the post-reform violation of parallel trends to be as large as a multiple $\bar{M}$ of the largest pre-reform violation. The formality effect remains bounded away from zero only for small departures, up to about $\bar{M} = 0.1$, and the self-employment effect only under near-exact parallel trends, whereas the employment effect includes zero even under the smallest departure. The compositional results are thus robust to conventional inference but sensitive to modest trend violations, which reinforces my decision to emphasize the wage null, the one result that is both precisely estimated and replicated out of sample.

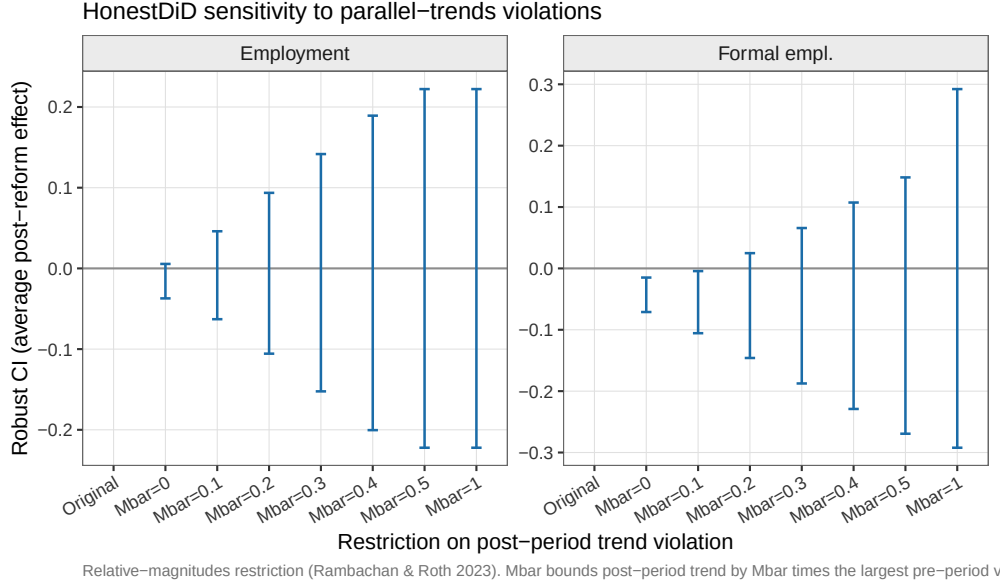


Figure 6: Sensitivity to violations of parallel trends (Rambachan and Roth, 2023). Robust confidence sets for the average post-reform effect under the relative-magnitudes restriction, which bounds the post-reform trend violation by $\bar{M}$ times the largest pre-reform violation.

7.5 Placebo reforms and leave-one-out

I assign fake reforms to each interior quarter of the pre-reform window and re-estimate the DiD using only pre-reform data. The wage placebos are uniformly null, the formality and self-employment placebos are mostly null with one marginal case, but the employment placebos are significant, which is a direct symptom of the pandemic-driven pre-trend in that outcome. Leave-one-department-out and leave-one-quarter-out re-estimation leave the point estimates essentially unchanged for all four outcomes, indicating that no single region or quarter drives the results. These checks are reported in the appendix.

7.6 Out-of-sample validation: the 2025 reform

Finally, I replicate the entire design around the second reform, which raised the minimum wage to 1,130 soles on 1 January 2025, using the 2024 and 2025 quarters. This reform is of almost identical magnitude to the 2022 one but occurs in a calmer, post-pandemic environment. Column (6) of Table 4 reports the estimates. The wage effect is again a null, which strengthens my central finding considerably: across two independent reforms in different macroeconomic conditions, the minimum wage did not raise the relative wages of low-skilled workers. The compositional effects, however, do not replicate: the formality and employment effects are null around the 2025 reform, and the self-employment effect reverses

sign. I take this seriously. It implies that the reallocation I document around 2022 is specific to that episode and may reflect the unusual dynamics of the post-pandemic labor market rather than a stable structural response to the minimum wage. The one conclusion that survives out of sample is the wage null.

8 Heterogeneity and distributional effects

8.1 Who adjusts?

Figure 7 reports the low- versus high-skilled DiD estimate for each headline outcome within demographic and regional subgroups. Three patterns stand out and are consistent with economic intuition about where the minimum wage binds hardest.

First, the wage null is uniform. There is no subgroup, whether defined by sex, age, area, or regional bite, in which the reform raised relative wages. The absence of a wage response is a pervasive feature of the data rather than an average that masks offsetting effects.

Second, the compositional adjustment is concentrated among the young and among women. The decline in formal employment and the rise in self-employment are largest for workers aged fourteen to twenty-five and for women, the groups whose wages are most likely to sit near the floor and whose attachment to formal employment is most tenuous. For workers over forty-five the effects are close to zero. This mirrors the emphasis on youth in the earlier Peruvian literature (Céspedes, 2005; Jaramillo and López, 2006) and the emphasis on women and the least skilled in the developing-country evidence (Katzkiewicz et al., 2021; Lemos, 2009).

Third, the employment-level effect, to the extent it exists, is concentrated among youth and in urban areas, but the subgroup estimates share the fragility of the aggregate and I do not emphasize them.

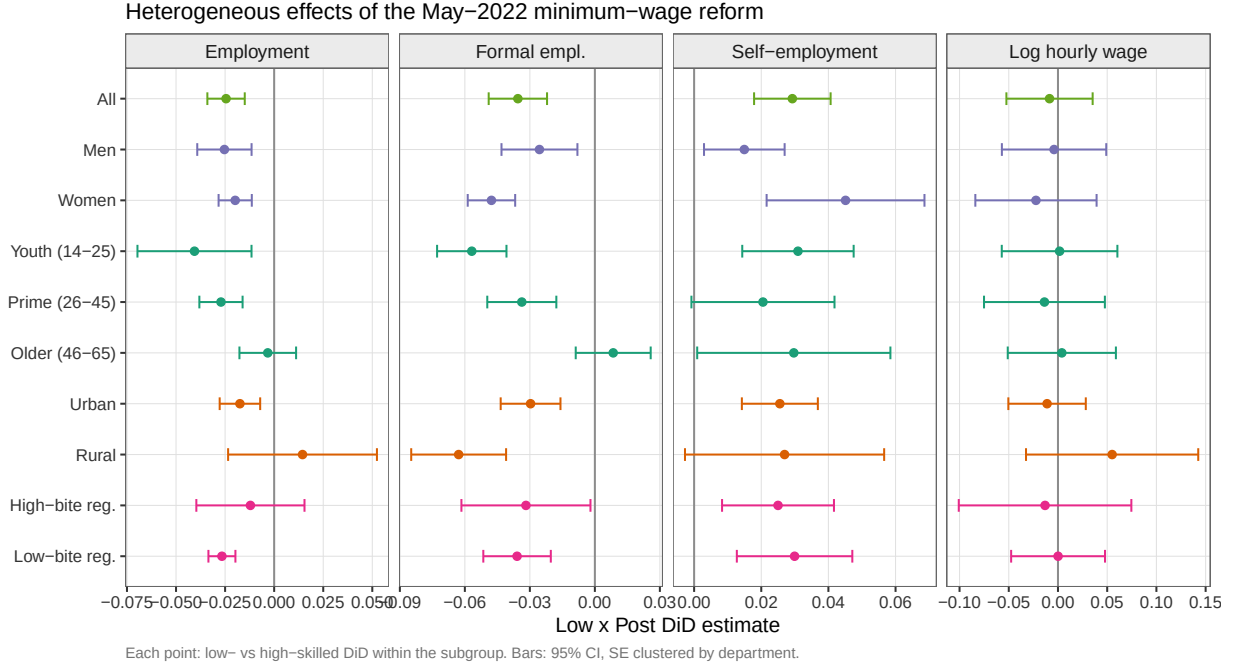


Figure 7: Heterogeneous effects of the 2022 reform. Each point is the $\text{Low} \times \text{Post}$ DiD estimate within the indicated subgroup, with ninety-five percent confidence intervals based on standard errors clustered by department.

8.2 Distributional wage effects

The absence of an average wage effect could in principle conceal offsetting movements at different points of the wage distribution, since a minimum wage is expected to compress the lower tail. To examine this I estimate unconditional quantile (recentered influence function) regressions in the manner of Firpo et al. (2009), applying the DiD specification to the influence function of each decile of the log real hourly wage. Figure 8 plots the $\text{Low} \times \text{Post}$ coefficient across deciles. The estimates are small and statistically insignificant throughout the distribution, including at the lower deciles where a binding minimum wage would be expected to raise wages. The distributional evidence therefore reinforces the headline null: the reform did not compress the lower tail of the low-skilled wage distribution relative to the high-skilled. This is consistent with the earnings distribution in Figure 5, which shows little movement toward the new floor, and with the interpretation that in a labor market with widespread non-compliance the statutory minimum did not bind for most low-skilled workers.

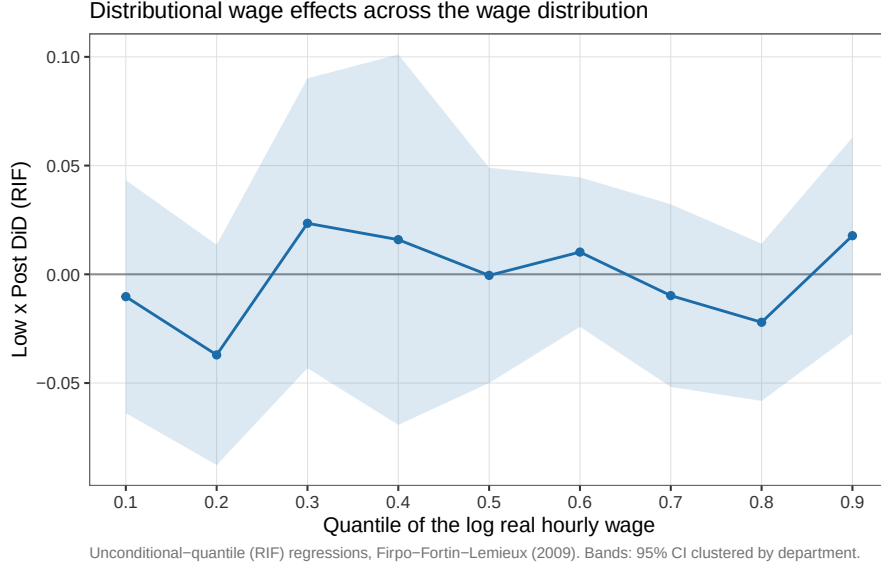


Figure 8: Distributional wage effects. Each point is the $\text{Low} \times \text{Post}$ DiD coefficient from an unconditional-quantile (RIF) regression at the indicated decile of the log real hourly wage (Firpo et al., 2009). Bands are ninety-five percent confidence intervals clustered by department.

9 Discussion

9.1 Mechanisms

The picture that emerges is of a minimum wage that, in the Peruvian setting, has little purchase on the pay of the workers it is meant to protect. The wage null is my most robust result, and it holds across two reforms, across the wage distribution, and across demographic groups. Two features of the institutional environment make this intelligible. The first is inflation. The 2022 increase of ten percent in nominal terms coincided with consumer price inflation of roughly eight percent, so the real value of the floor barely moved and then eroded. The second, and more fundamental, is informality. The overwhelming majority of low-skilled Peruvian workers are informal or self-employed, and for them the statutory minimum is not enforced. Compliance is high only within registered formal firms, which employ a minority of the low-skilled. A wage floor that reaches only a small, selected part of the relevant labor force cannot be expected to move the average wage of that force by much, and the earnings distribution in Figure 5 confirms that it did not.

If the reform did not raise wages, the natural question is whether it had any effect at all. Around the 2022 reform I find evidence of a shift from formal employment toward self-employment, concentrated among the young and among women. The most plausible

mechanism is the classic one of a binding floor in the covered sector inducing some reallocation to the uncovered sector, the displacement channel emphasized by Jaramillo and López (2006) for Peru and by Ham (2018) for Honduras. The rise in hours in the main occupation is consistent with movement toward self-employment, where hours are less constrained. I am cautious, however, because this compositional effect does not replicate around the 2025 reform. The reversal suggests that at least part of what I observe around 2022 reflects the turbulent, uneven recovery of the post-pandemic labor market rather than a structural response to the minimum wage. I therefore present the reallocation as a documented feature of the 2022 episode rather than as a general law.

9.2 Reconciliation with the previous literature

My findings both agree and disagree with the prior evidence, and the pattern of agreement is informative. They agree with the older Peruvian work of Jaramillo and López (2006), who concluded that the 2003 reform was largely innocuous and found displacement toward informality, and with Céspedes (2005), who found effects concentrated among the young. They agree with the broad international consensus that minimum wage increases in the range studied produce small employment effects (Card and Krueger, 1995; Cengiz et al., 2019).

They disagree, however, with the most recent Peruvian study. Jiménez (2023) finds that the 2016 reform raised formal wages by four to five percent and increased formality, a lighthouse pattern that I do not reproduce for 2022. Several differences plausibly account for the divergence. The 2016 reform occurred in a low-inflation period, so its real bite was larger and more durable than that of the inflation-eroded 2022 increase. The identification differs: Jiménez (2023) uses a department-level dose defined on the formal wage distribution and estimates effects for the department, whereas I contrast low- and high-education workers at the individual level, a design that places more weight on the large informal segment of the low-skilled. And the periods differ fundamentally: 2016 was a period of stable growth, while 2022 was a post-pandemic recovery. The contrast suggests that the labor-market effects of the Peruvian minimum wage are state-dependent, larger and more lighthouse-like when the real increase is substantial and the economy is stable, and muted when inflation erodes the increase and informality absorbs the adjustment.

9.3 External validity and limitations

Several limitations bound the interpretation of my estimates. First, identification rests on the assumption that, absent the reform, low- and high-education workers would have followed parallel paths. I have tested this where possible and bounded the consequences of its failure, but I cannot rule out differential shocks correlated with education, particularly in the volatile 2021–2022 window. This is why I lean on the wage result, whose pre-trends are flat and which replicates out of sample, and treat the employment-level result as inconclusive. Second, the design identifies effects on low-skilled workers relative to high-skilled workers. If the reform also affected high-skilled workers, for example through general-equilibrium spillovers or ripple effects up the wage distribution, my estimates understate or overstate the absolute effect on the low-skilled. I view large spillovers to the tertiary-educated as unlikely given that their median wage sits well above the floor, but I cannot exclude them. Because the comparison group is not entirely unexposed, any bias from imperfect exposure attenuates my estimates toward zero, so the wage null is if anything a conservative reading. Third, the data are repeated cross-sections, so I cannot follow individuals across the reform and cannot separate changes in the composition of the employed from changes in individual outcomes. Fourth, informality for 2024 is measured with a reconstructed indicator rather than the official one; my robustness checks suggest this does not drive the results, but it introduces measurement error. Finally, my window covers at most eleven post-reform quarters, so I speak to short- and medium-run effects and cannot address the long-run adjustment of capital or firm entry and exit.

9.4 Policy implications

The results carry a sober message for minimum wage policy in a high-informality economy. A statutory floor that is not enforced for the majority of its intended beneficiaries does little to raise their pay, and to the extent it binds in the formal sector it may push some workers toward less protected arrangements. Raising the minimum wage is therefore an imperfect instrument for improving the earnings of the low-skilled in Peru, a conclusion that echoes Jaramillo and López (2006). This does not imply that the minimum wage is harmless or useless; my estimates are relative and my window is short. But it does suggest that complementary policies aimed at formalization and enforcement are prerequisites for the minimum wage to reach the workers it is designed to protect. Absent such policies, the main margin of adjustment is the composition of employment rather than the level of pay.

10 Conclusion

I have studied how Peru’s 2022 minimum wage increase, from 930 to 1,025 soles, affected workers with different levels of education, using sixteen quarters of the ENAHO employment survey and a difference-in-differences design that exploits the sharply different exposure of low- and high-education workers to a single national reform. My central finding is a precisely estimated null: the reform did not raise the real wages of low-education workers relative to high-education workers, at the mean or anywhere in the distribution. This conclusion is robust to a wide range of specifications and, importantly, replicates around the subsequent reform of January 2025. Around the 2022 reform I also find evidence of a shift from formal employment toward self-employment, concentrated among the young and among women, though this compositional effect is less robust and does not carry over to the 2025 episode. The overall employment effect cannot be pinned down and is confounded by the uneven post-pandemic recovery.

The most economical reading of these results is that, in a labor market where informality is pervasive and enforcement is weak, a moderate and inflation-eroded increase in the statutory minimum does little to lift the pay of the low-skilled and instead operates, when it operates at all, on the margin between formal and informal work. This qualifies the light-house view of the Peruvian minimum wage that emerges from studies of earlier reforms and points to the informal margin as the central channel of adjustment.

Two directions for future research follow. The first is to exploit the panel component of ENAHO, where available, to follow individuals across reforms and to distinguish compositional from individual-level effects. The second is to study the interaction between minimum wage policy and formalization policy directly, since my results suggest that the returns to raising the floor depend on whether the floor can be made to bind. As additional post-reform quarters accumulate, it will also become possible to trace the longer-run adjustment that a short window such as mine cannot capture.

Declarations

Declarations of interest: none.

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Data availability: the ENAHO microdata are publicly available from the Instituto Nacional de Estadística e Informática (INEI). The code that reproduces every table and figure is available in the online replication repository.

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A Data appendix

A.1 Sample and variable construction

I pool the twenty quarterly ENAHO employment modules (module 500) for 2021Q1–2025Q4; the main analysis uses 2021Q1–2024Q4 and the validation uses 2024Q1–2025Q4. The working-age sample comprises individuals aged 14 to 65 with non-missing education. All statistics use the module employment weight (`fac500a`). Departments are the 25 first-level administrative units derived from the first two digits of `ubigeo`; the urban indicator is derived from the sampling stratum.

Education and skill groups. Educational attainment is ENAHO variable `p301a`. I verified its coding against the weighted attainment distribution. Low-skilled workers are those with `p301a` in 1–6 (no education through complete secondary); high-skilled workers are those with `p301a` in 7–11 (non-university higher through postgraduate). Special education (code 12) and missing values are excluded.

Labor income and wages. For dependent employees, monthly labor income in the main occupation is the reported gross pay including overtime (`p524a1`); for own-account workers it is net profit (`p530a`). I deflate by the Lima consumer price index to December 2021 soles. The real hourly wage divides monthly labor income by 4.345 times weekly hours in the main occupation (`p513t`, imputed where missing). Wages are winsorized at the 0.5th and 99.5th percentiles. As a robustness check I also construct the imputed, deflated, annualized labor-income aggregate used by Jiménez (2023), namely the sum of `i524a1`, `d529t`, `i530a`, and `d536` divided by twelve; results are unchanged.

Informality. INEI’s official informal-employment indicator (`ocupinf`, coded 1 for informal and 2 for formal in these files) is present for 2021–2023 but absent for 2024–2025. I reconstruct a consistent indicator from primitives available in every year. A wage employee or domestic worker is classified as formal if affiliated to a pension system (`p558a1`–`p558a4`) or holding a formal labor contract (`p511a` in 1–6); an employer or own-account worker is classified as formal if the productive unit is registered (`p510a1`); unpaid family workers are always informal. Validated against `ocupinf` on the overlapping years 2021–2023, this rule agrees with the official classification for 88.3 percent of employed workers and reproduces the official informality rate to within about two and a half percentage points. I use the official indicator wherever available and the reconstruction only for 2024–2025.

Minimum wage and bite. The statutory minimum wage schedule is taken from the decrees and verified against the Banco Central de Reserva del Perú series (PN02124PM). The department Kaitz index is the ratio of the post-reform minimum (1,025 soles) to the department’s pre-reform (2021Q1–2022Q1) weighted median monthly wage of dependent employees. The fraction-affected measure is the pre-reform weighted share of dependent employees earning below 1,025 soles.

A.2 Supplementary results

Table 5 reports the in-time placebo reforms and confirms that the wage placebos are null while the employment placebos are contaminated by the pandemic pre-trend. The doubly robust estimates of Sant’Anna and Zhao (2020) appear alongside the two-way fixed-effects estimates in Table 2. Figure 9 shows the event study around the January-2025 reform used in the out-of-sample validation.

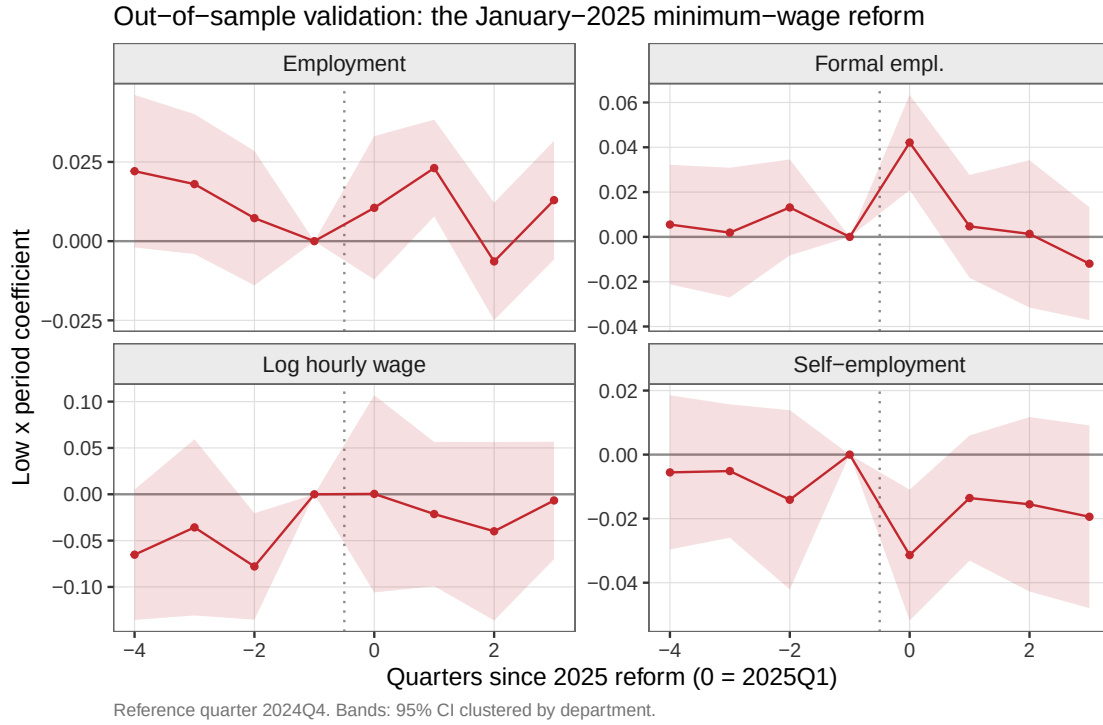


Figure 9: Event study around the January-2025 reform (out-of-sample validation). Coefficients on the interaction of the low-skill indicator with quarter indicators, reference quarter 2024Q4. Bands are ninety-five percent confidence intervals clustered by department.

Table 5: In-time placebo reforms (pre-reform window only)

Outcome	Placebo 2021Q2	Placebo 2021Q3	Placebo 2021Q4
Log hourly wage	0.020	-0.018	-0.005
Employment	-0.057***	-0.034***	-0.026***
Formal empl.	-0.021*	-0.012	-0.005
Self-employment	0.022*	-0.009	0.007

Notes: Each entry is the $\text{Low} \times \text{Post}$ coefficient from a placebo DiD that assigns a fake reform to the indicated quarter, estimated using only pre-reform data (2021Q1–2022Q1). Standard errors clustered by department. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.